



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO KUMAR FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

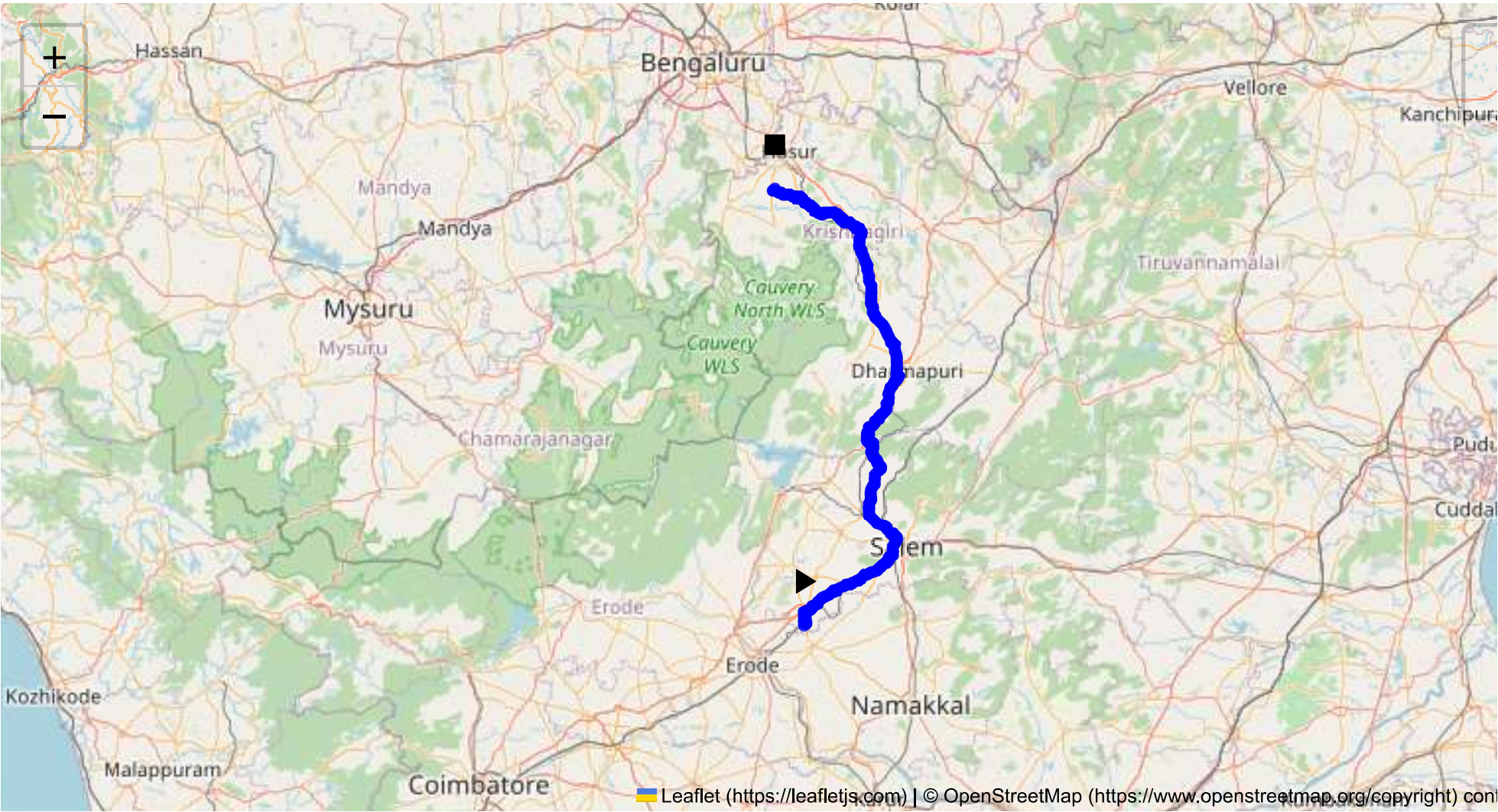
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 182.46 km
Estimated Duration: 3.6 hours
Adjusted Duration (Heavy Vehicle): 4.5 hours
Start: (11.4381, 77.8734)
End: (12.627831, 77.785929)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri, Tamil Nadu to Krishnagiri, via Hosur, spans approximately 182 kilometers. It typically involves traversing National Highways like NH 44 and NH 44 variants, known for connecting major towns and industrial regions. This is a commonly used route due to its direct path and established infrastructure.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu experiences a tropical climate, with hot summers, a significant rainy season from June through September (monsoon), and a relatively mild winter. Monsoon season could result in heavy rainfall, leading to slippery roads and potential flooding, especially on poorly drained sections. Visibility might be reduced during heavy rains, adding to the risk factor for drivers of heavy vehicles.

3. Analysis of Traffic Patterns

Traffic along this route can be intense, especially nearing urban areas like Salem and Dharmapuri, with most congestion occurring during peak hours from 8:00 AM to 10:00 AM and 5:00 PM to 8:00 PM. Weekends might also see increased traffic due to local events or tourist flows.

4. Assessment of Road Quality and Infrastructure

The quality of roads on major national highways such as NH 44 is generally good, with regular maintenance ensuring smooth surfaces. However, secondary roads or connectors may have varying quality, with potholes and uneven surfaces, especially post-monsoon.

5. Suggestions for Alternative Routes for Emergencies

In case of roadblocks or emergencies, an alternative route could be via NH 79 and then connecting to smaller highways towards Krishnagiri. These alternatives may lengthen travel time slightly but could be necessary during adverse conditions.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu follows central Indian laws regarding the transport of hazardous materials, requiring proper documentation and labeling. Movement during certain hours might be restricted in urban areas to prevent accidents. Trucks carrying hazardous materials often need to follow specific routes designed to minimize risk in residential or crowded zones.

7. Overview of Historical Incidents

There have been incidents in the past involving heavy vehicles, including overturns in the Salem district due to sharp curves and sudden narrow road sections. Transport of hazardous materials needs heightened vigilance, often necessitating night travel restrictions.

8. Environmental Considerations and Sensitive Areas

The route passes close to some sensitive environmental zones, including forested areas as you approach Dharmapuri. There should be strict compliance with noise and pollution norms to protect these habitats.

9. Analysis of Communication Coverage

Most urban and high-traffic rural areas offer good mobile coverage. However, some rural segments between major towns can experience network coverage issues, affecting real-time communication and GPS use.

10. Estimated Emergency Response Times

Emergency response times can vary; in urban areas like Salem, it is faster (about 15-25 minutes), but in more remote parts, it might take up to an hour for services such as medical aid or towing services to reach the location.

11. Overall Summary of Risk Assessment

Overall, while the route is manageable with good infrastructure on major stretches, several risk factors must be managed:

- Weather conditions and road upkeep post monsoon.
- Traffic congestion and varying road quality on less major routes.
- Regulatory compliance concerning hazardous material transport.
- Potential communication dead zones in rural areas. With adequate preparation, such as planning around peak traffic and ensuring route familiarity, these risks can be mitigated effectively. Keeping local emergency contacts and alternative routing in mind is advisable for enhancing route safety.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	12.60628, 77.86014	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	High	11.44900, 77.87406	15 KM/Hr
4	Turn	High	11.45028, 77.87202	15 KM/Hr
5	Turn	Medium	11.46400, 77.87323	30 KM/Hr
6	Turn	Medium	11.46765, 77.87037	30 KM/Hr
7	Turn	High	11.47403, 77.86841	15 KM/Hr
8	Turn	Medium	11.47986, 77.88397	30 KM/Hr
9	Turn	High	11.49149, 77.89601	15 KM/Hr
10	Turn	High	11.49224, 77.89601	15 KM/Hr
11	Turn	Medium	11.53512, 77.95717	30 KM/Hr
12	Turn	Medium	11.53606, 77.95869	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
13	Turn	Medium	11.62403, 78.10893	30 KM/Hr
14	Turn	Medium	11.62932, 78.11659	30 KM/Hr
15	Turn	High	11.64183, 78.11810	15 KM/Hr
16	Turn	Medium	11.64338, 78.11998	30 KM/Hr
17	Turn	Medium	11.67903, 78.12796	30 KM/Hr
18	Turn	Medium	11.69532, 78.10764	30 KM/Hr
19	Turn	Medium	11.73968, 78.05098	30 KM/Hr
20	Turn	Medium	11.86790, 78.08347	30 KM/Hr
21	Turn	Medium	11.91357, 78.05868	30 KM/Hr
22	Turn	Medium	11.92556, 78.06499	30 KM/Hr
23	Turn	Medium	11.93296, 78.06432	30 KM/Hr
24	Turn	Medium	11.93535, 78.06150	30 KM/Hr
25	Turn	Medium	11.95531, 78.04698	30 KM/Hr
26	Turn	High	11.95803, 78.04964	15 KM/Hr
27	Turn	Medium	11.96068, 78.04960	30 KM/Hr
28	Turn	High	11.96370, 78.05142	15 KM/Hr
29	Turn	High	11.96542, 78.04938	15 KM/Hr
30	Turn	Medium	11.96650, 78.04927	30 KM/Hr
31	Turn	Medium	11.96890, 78.05109	30 KM/Hr
32	Turn	Medium	12.12134, 78.13243	30 KM/Hr
33	Turn	Medium	12.50466, 78.03128	30 KM/Hr
34	Blind Spot	Blind Spot	12.51373, 78.03208	10 KM/Hr
35	Turn	Medium	12.51382, 78.02925	30 KM/Hr
36	Turn	Medium	12.51628, 78.02574	30 KM/Hr
37	Turn	High	12.52225, 78.02370	15 KM/Hr
38	Turn	Medium	12.52596, 78.01126	30 KM/Hr
39	Turn	High	12.55119, 77.97237	15 KM/Hr
40	Turn	Medium	12.55300, 77.96561	30 KM/Hr
41	Turn	Medium	12.56511, 77.95333	30 KM/Hr
42	Turn	Medium	12.56244, 77.92201	30 KM/Hr
43	Turn	Medium	12.56400, 77.91182	30 KM/Hr
44	Turn	Medium	12.57383, 77.89811	30 KM/Hr
45	Turn	Medium	12.58098, 77.89496	30 KM/Hr
46	Turn	Medium	12.58581, 77.88072	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
47	Turn	Medium	12.59024, 77.87703	30 KM/Hr
48	Turn	Medium	12.59225, 77.86934	30 KM/Hr
49	Turn	Medium	12.60278, 77.86253	30 KM/Hr
50	Turn	Medium	12.60622, 77.84708	30 KM/Hr
51	Turn	Medium	12.60972, 77.84299	30 KM/Hr
52	Turn	Medium	12.60976, 77.83567	30 KM/Hr
53	Turn	High	12.62538, 77.78575	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
5	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
6	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
8	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
9	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
10	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
12	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
17	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
18	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
19	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
21	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
22	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
23	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
24	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
25	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium
26	hospital	Global Medical Centre Jagirammapalayam	11.6894409, 78.1108847	30 km/h	Medium
27	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium
31	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
32	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
35	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
36	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
37	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
11	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
13	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium